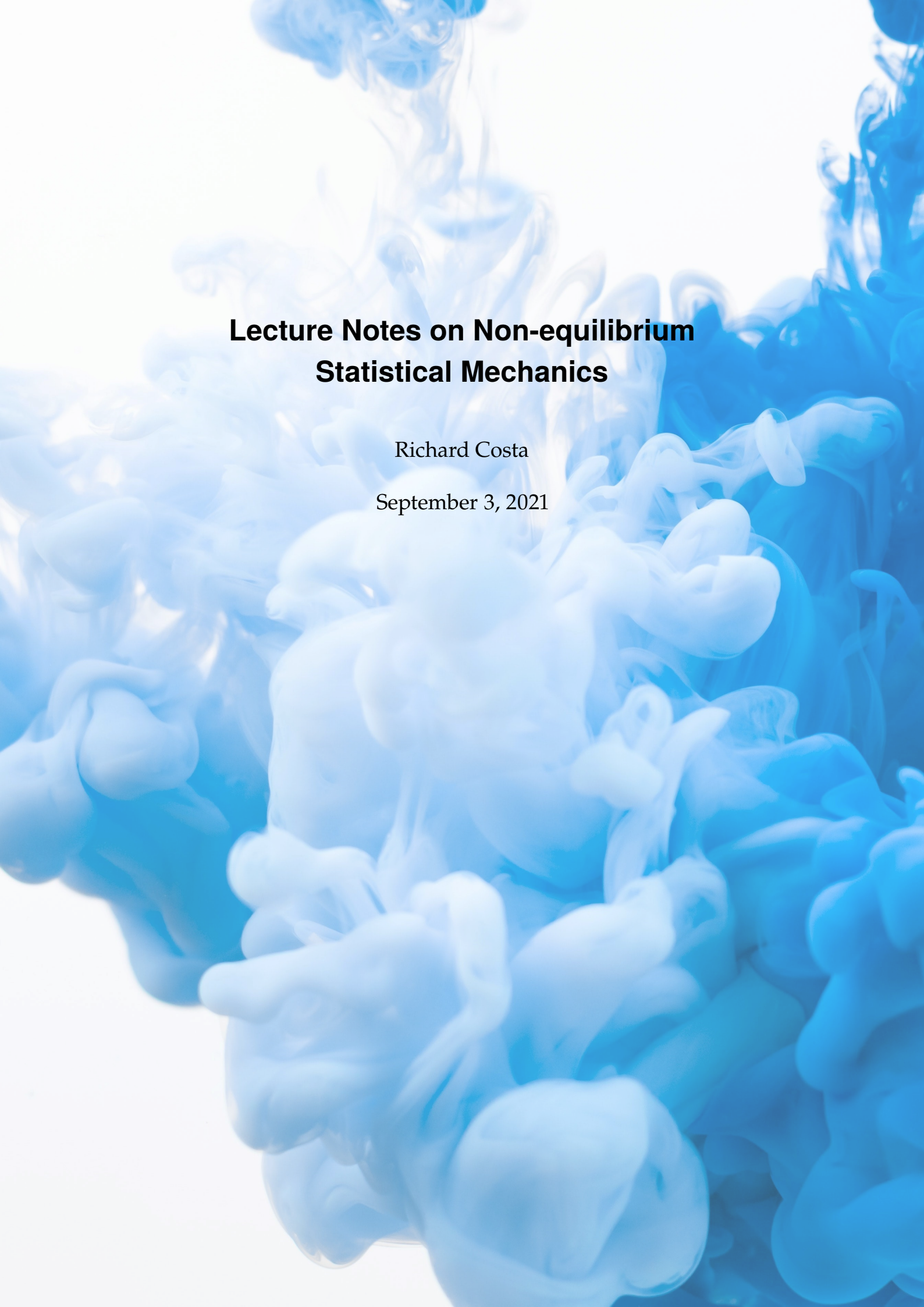

Lecture Notes on Non-equilibrium Statistical Mechanics

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1.1 Introduction

In a first course in statistical mechanics we usually focus on systems at thermodynamic equilibrium. In those systems the macroscopic state does not depend on time, i.e., everything that was supposed to happen regarding its thermal evolution already happened and the system reached equilibrium.

The theoretical framework used to describe these kind of systems is well established in the theory of ensembles, specifically the microcanonical, canonical and grand canonical ensemble. in this context we have already studied several techniques to obtain the partition function and, from it, to obtain the thermodynamics we are interested in, whether the classical case or the quantum one.

In these notes we are going to study systems out of equilibrium. The description of those kind of systems is far less complete and satisfactory than those in equilibrium, so it is a theory still under development.

In practice there are several ways to describe a system out of equilibrium, some being better than others depending on the type of problem at hand. However, there are two main ways of describing OFQ [1] systems: through kinetic theories and stochastic theories.

1: from now on, when convenient, OFQ stands for *out of equilibrium*.

Kinetic theories make use of something called *coarse graining*, which can be understood as a way to systematically use approximations in order to describe the system, controlling how much information we need - or want - to add to our model and, therefore, controlling the complexity of it and at the same time being sure of what you are leaving out. An analogy would be expanding some function into Taylor series and truncating it according to what the situation requires of us. Examples of kinetic theories and coarse graining descriptions include

- *The Boltzmann Equation*. This equation gives us the time evolution of the position and velocity **distributions** of particles (from a gas) and it is more appropriate to describe diluted systems, where interactions are short range in nature. We will talk more about this equation soon.

- *The Vlasov Equation*. Used to describe quasi-neutral [2] plasma and more appropriate for long-range interactions (in this case the eletromagnetic interaction).

2: Quasi-neutrality means that when we see the system as a whole there is electric charge neutrality, that is, the net charge is zero. But locally we have regions that are not neutral.

Stochastic theories allows to construct probabilistic models, usually inspired in the Brownian movement. In this realm we will see the Langevin Equation, the Fokker-Planck equation and the master equation with some applications.

Now, our entry point will be the Boltzmann equation, and in these lectures we will see two ways of deriving it. One is more phenomenological, and the other more formal [3] . We will start with the former, so we can have more physical insight into what is going on.

3: The formal derivation starts with the Liouville equation for a system with many particles (for example 10^{23}). We then work with these equations and get to the point where we have a hierarchy of equations denoted *BBGKY Hierarchy*. Both the Boltzmann equation and the Vlasov equation can be obtained from this hierarchy, but we'll see them in due time.

To arrive at the Boltzmann equation we start with the following assumptions:

1. The system is a dilute gas.
2. Particles interact only during collisions.
3. (for simplicity) The gas is composed of only one type of particle, each of which with mass m.
4. The system is finite with ideal walls which reflect particles as if in an elastic collision. We can also consider a system with an infinite number of particles, but we may require that in the thermodynamic limit the density is finite and constant N/Vc.
5. The system is under a force $\vec{F}(\vec{r}, t)$ which, for simplicity, we assume it does not depend on the velocity of particles [4] .

4: For the Lorentz force this assumption does not hold, since this force depends on the velocity of particles.

Let us expand on some of those assumptions:

First assumption. For a dilute gas we can think that we're dealing with a system of low density. More formally, the gas is dilute when the average distance between particles is much greater than their sizes: $d \gg \lambda$. First, let v be the specific volume of the gas. Then

$$\begin{aligned} v &= \frac{V}{N} \\ &= \left(\frac{N}{V}\right)^{-1} = n^{-1} \end{aligned}$$

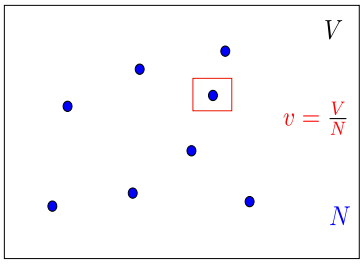

Figure 1.1: Representation of a gas made up of N particles in a volume V

The cubic distance between particles is proportional to n, so

$$\begin{aligned} d &\sim v^{\frac{1}{3}} \\ &\sim n^{-\frac{1}{3}} \end{aligned}$$

We can describe this distance with the De Broglie wavelength

$$\lambda_B = \frac{h}{p} = \frac{h}{\sqrt{2mE}}.$$

Recall that $\langle E \rangle = 3kT/2$, then

$$\lambda_B = \frac{h}{\sqrt{3mkT}}.$$

So the dilute gas condition ($d \gg \lambda_B$) can be rewritten as

$$n^{-\frac{1}{3}} \gg \frac{h}{\sqrt{3mkT}}.$$

We can also characterize this gas condition in terms of the *thermal wavelength* $\lambda = h/(\sqrt{2\pi mkT})$. Since λ is of the same order as λ_B then $n^{-\frac{1}{3}} \gg \lambda$. Therefore, our first assumption can be translated as

$$\lambda_B^3 n \ll 1, \text{ or } \lambda^3 n \ll 1.$$

We can use this condition to characterize the dilute gas as well as the classical regime.

Second assumption. Since we have a dilute gas or low density of particles, those particles move as free particles. Only on short time intervals they interact with each other, that is, they collide. The nature of collisions is determined by what is called *cross section*, which will see in more detail later on.

1.2 Phase space and the distribution function

This space [5] has 6 dimensions, being 3 the position coordinates and 3 the momenta coordinates, for each particle/molecule. A point in this space represents the dynamic state of a particle/molecule. For N particles or molecules, the state of the system is represented by N points in this space. The state is completely determined by the position and momenta coordinates.

5: Also denoted *μ space.*

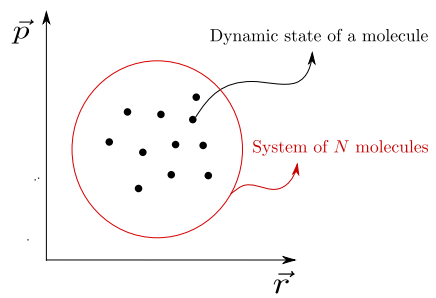

Figure 1.2: Phase Space.

We now define a *distribution function* f in this space such that $f(\vec{r}, \vec{p}, t)d^3d^3p$ is the number of particles in time t that have are in an element of position volume d^3r around $\vec{r}$ and momenta volume d^3p around $\vec{p}$.

The notation $d^3 \equiv dr_1dr_2dr_3 \equiv r_xr_yr_z$, and the same for the momenta. [6]

We suppose that $f(\vec{r}, \vec{p}, t)$ is a continuous function in its arguments. This isn't really true, but since there are usually a huge number of particles in the volume element d^3rd^3p and the density of points vary smoothly, we make this assumption [7] .

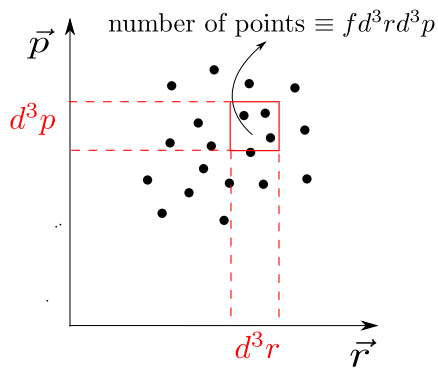

Figure 1.3: Distribution function and number of points in phase space.

The goal of Kinetic Theory is to find the distribution function f. This function depends on the problem at hand: on the type of interactions between the particles, initial conditions, boundary conditions, etc.

The distribution function can be determined by the Boltzmann Equation and so when we solve this equation we are effectively finding what the distribution function is. With the distribution function we can then find the ensemble average of the physical quantities we are interested in. For instance, let $\chi(\vec{r}, \vec{v}, t) = m\vec{v}^2/2$. Then, its average in time t and position $\vec{r}$ is obtained by integration over velocities:

$$\langle \chi(\vec{r}, t) \rangle = \frac{\int d^3v \chi(\vec{r}, \vec{v}, t) f(\vec{r}, \vec{v}, t)}{\int d^3v f(\vec{r}, \vec{v}, t)},$$

6: Sometimes we will write this and other functions as functions of the velocity $\vec{v}$ instead of momentum $\vec{p}$.

7: A more rigorous way would be to consider that f describe point particles and so can be described as a sequence of Dirac deltas.

where we assumed that f is already normalized. The denominator is the number of particles in the volume element $d^3r d^3v$ of the phase space. We may denote

$$n(\vec{r}, t) = \int d^3v f(\vec{r}, \vec{v}, t)$$

so $n(\vec{r}, t)$ is the number of particles per unit volume around $\vec{r}$ at time t.

1.3 Deriving the Boltzmann Equation

Let $d^3r d^3v$ be the volume element in the μ space of our system. Now, suppose there are no collisions between particles. In a instant in time $t' = t + dt$ the molecules will move due to the external force and end up in another position as in the figure below:

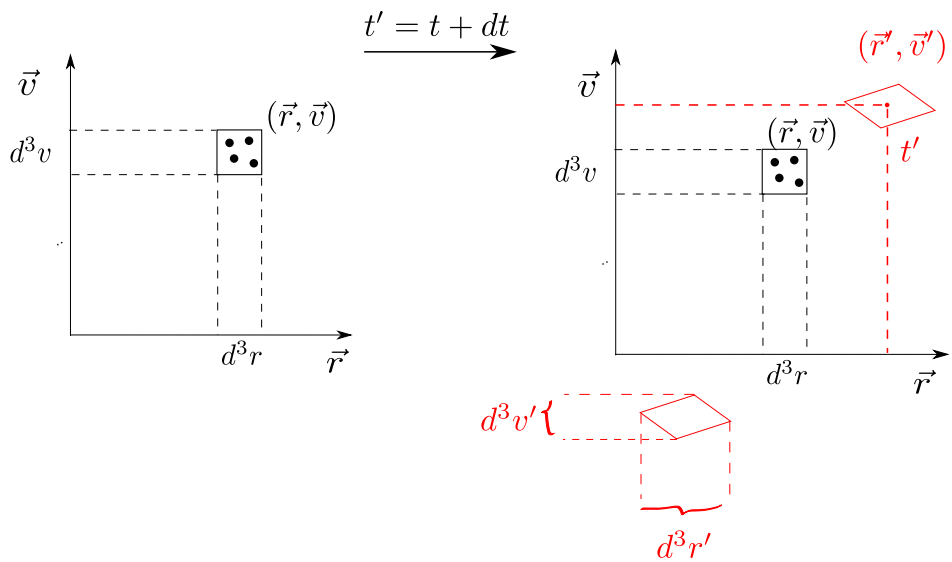

Figure 1.4: Volume elements under external force at a later instant in time.

The $d^3r d^3v$ volume element will be displaced to the coordinates $(\vec{r}', \vec{v}')$. This moviment will be determined by the initial conditions and by the force applied to the system. That way, we may write

$$\begin{aligned} \vec{r}' &= \vec{r} + \dot{\vec{r}} dt = \vec{r} + \vec{v} dt, \\ \vec{v}' &= \vec{v} + \dot{\vec{v}} dt = \vec{v} + \frac{\vec{F}}{m} dt. \end{aligned} \tag{1.1}$$

We can show that both volume elements are equal, that is, $d^3r d^3v = d^3r' d^3v'$. As a matter of fact, the relationship between the two is given by

$$d^3r' d^3v' = |J| d^3r d^3v,$$

where J is the Jacobian of the transformation between the coordinates $(\vec{r}, \vec{v})$ and $(\vec{r}', \vec{v}')$. At first order in time the Jacobian is equal to 1 and then the volume elements are qual. This is showed in the exercises section.

All molecules that were initially at $d^3r d^3v$ will be at $d^3r' d^3v'$. This is due to the facts of our situation: initial conditions of very close particles in a volume element are similar, all particles are subjected to the same external force and there are no collisions. Since all molecules move to the new volume element, the distribution functions are the same:

$$f(\vec{r}, \vec{v}, t) d^3r d^3v = f(\vec{r}', \vec{v}', t) d^3r' d^3v' \implies f(\vec{r}, \vec{v}, t) = f(\vec{r}', \vec{v}', t).$$

Using (1.1) we can write

$$f(\vec{r}, \vec{v}, t) = f\left(\vec{r} + \vec{v}dt, \vec{v} + \frac{\vec{F}}{m}dt, t + dt\right).$$

When there are collisions not all molecules from $d^3r d^3v$ will end up at $d^3r' d^3v'$, so we will analyze what happens in this case. We will find that there will be a term stemming from collisions which, fairly enough, we will call it the *collision term*.

Now suppose we have molecules that do collide. Consider molecules that have similar velocities and under the same external force. Due to the collision there will be a significant change in their positions and velocities. They may not end up in the volume $d^3r' d^3v'$ centered at $(\vec{r}', \vec{v}')$. Furthermore, we may have molecules that initially were at a different volume element (one not centered in $(\vec{r}, \vec{v})$) but ended up in $d^3r' d^3v'$ anyway. So

$$f(\vec{r}, \vec{v}, t) \neq f\left(\vec{r} + \vec{v}dt, \vec{v} + \frac{\vec{F}}{m}dt, t + dt\right).$$

This difference of the distributions functios due to collisions is what we call the collision term. It is a constant and we define it [8] for now by

$$\underbrace{\left(\frac{\partial f}{\partial t}\right)_{\text{Coll.}}}_{\equiv C} dt.$$

Therefore,

$$f\left(\vec{r} + \vec{v}dt, \vec{v} + \frac{\vec{F}}{m}dt, t + dt\right) - f(\vec{r}, \vec{v}, t) = \left(\frac{\partial f}{\partial t}\right)_{\text{coll}} dt \tag{1.2}$$

Now, expanding the first distribution function above in Taylor series [9] around the point $(\vec{r}, \vec{v}, t)$:

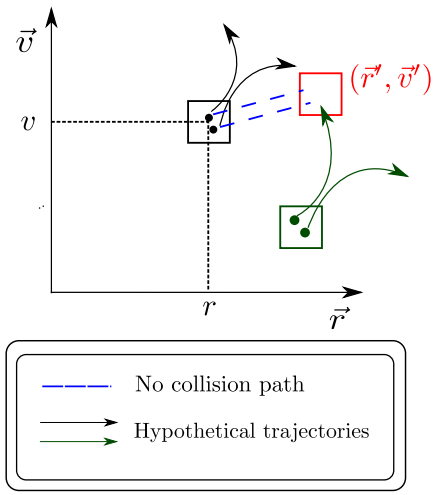

Figure 1.5: Possible collisions.

8: For historical reasons it is defined as a partial derivative. Other texts just call it C. Either way, note that we must have units of f on the right side of (1.2).

9: The Taylor series expansion at first order for n variables around the point $(a_1, \cdots, a_n)$ is given by

$$f(x_1, \cdots, x_n) \approx f(a_1, \cdots, a_n) + \sum_{i=1}^{n} \frac{\partial}{\partial x_i} f(a_1, \cdots, a_n)(x_i - a_i).$$

For more see here and here. In our case we have $6N$ variables, where $3N$ are for positions and $3N$ for momenta.

$$
\begin{aligned}
f\left(\vec{r}+\vec{v}dt, \vec{v}+\frac{\vec{F}}{m}dt, t+dt\right) &\approx f(\vec{r},\vec{v},t)+\sum_{i=1}^{3N}\frac{\partial}{\partial x_i}f(\vec{r},\vec{v},t)(\vec{r}+\vec{v}dt-\vec{r}) \\
&+\sum_{j=1}^{3N}\frac{\partial}{\partial v_j}f(\vec{r},\vec{v},t)\left(\vec{v}+\frac{\vec{F}}{m}dt-\vec{v}\right)+\frac{\partial}{\partial t}f(\vec{r},\vec{v},t)(t+dt-t) \\
&= f(\vec{r},\vec{v},t)+\sum_{i=1}^{3N}\frac{\partial}{\partial x_i}f(\vec{r},\vec{v},t)\vec{v}dt+\sum_{j=1}^{3N}\frac{\partial}{\partial v_j}(\vec{r},\vec{v},t)\frac{\vec{F}}{m}dt+\frac{\partial}{\partial t}f(\vec{r},\vec{v},t)dt \\
&= f(\vec{r},\vec{v},t)+\nabla_r f\cdot\vec{v}dt+\nabla_v f\cdot\frac{\vec{F}}{m}dt+\frac{\partial}{\partial t}f(\vec{r},\vec{v},t)dt,
\end{aligned}
$$

where we have kept only first order terms of the expansion. Replacing this into (1.2) we get

$$
\frac{\partial}{\partial t}f+\nabla_r f\cdot\vec{v}+\nabla_v f\cdot\frac{\vec{F}}{m}=\left(\frac{\partial f}{\partial t}\right)_{\text{Coll.}}.
$$

Now, since the scalar product of gradients of f with vectors is commutative, that is,

$$
\nabla_r f\cdot\vec{a}=\vec{a}\cdot\nabla_r f \text{ and } \nabla_v f\cdot\vec{a}=\vec{a}\cdot\nabla_v f
$$

for any vector $\vec{a}\in\mathbb{R}^n$, then we finally have the Boltzmann equation:

To see the commutation, write the scalar product explicitly. For example:

$$
\begin{aligned}
\nabla_r f\cdot\vec{a} &= \frac{\partial f}{\partial x}a_x+\frac{\partial f}{\partial y}a_y+\frac{\partial f}{\partial z}a_z \\
&= a_x\frac{\partial f}{\partial x}+a_y\frac{\partial f}{\partial y}+a_z\frac{\partial f}{\partial z} \\
&= \vec{a}\cdot\nabla_r f
\end{aligned}
$$

Boltzmann Equation

$$
\left(\frac{\partial}{\partial t}+\vec{v}\cdot\nabla_r+\frac{\vec{F}}{m}\cdot\nabla_v\right)(\vec{r},\vec{v},t)=\left(\frac{\partial f}{\partial t}\right)_{\text{Coll.}}. \tag{1.3}
$$

We can also write the Boltzmann equations in terms of momenta. Just note that $p_i=mv_i$, so

$$
\nabla_v f\cdot\frac{F}{m}=\frac{\vec{F}}{m}\cdot\nabla_v f=F_i\frac{1}{m}\frac{\partial f}{\partial v_i}=F_i\frac{\partial f}{\partial(mv_i)}=F_i\frac{\partial f}{\partial p_i}=\vec{F}\cdot\nabla_p.
$$

Therefore,

Boltzmann Equation

$$
\left(\frac{\partial}{\partial t}+\vec{v}\cdot\nabla_r+\vec{F}\cdot\nabla_p\right)f(\vec{r},\vec{p},t)=\left(\frac{\partial f}{\partial t}\right)_{\text{Coll.}}. \tag{1.4}
$$

However, note that we **do not** have an expression for the collision term just yet. We did nothing more than just state that there is a difference between the distribution functions and worked it out from there. We will now derive the expression for the collision term.

1.4 The Collision Term

First, note that the collision term is the (net) number of particles that get into a volume element due to collisions. Let us denote the number of particles that enter d^3rd^3v as $\overline{R}$ and the number of particles that escape d^3rd^3v as R. Therefore we can write

$$\left(\frac{\partial f}{\partial t}\right)_{\text{Coll.}} dt = (\overline{R} - R)dt.$$

Now, R is directly related to the number of collisions because we assume the original volume element is so small (infinitesimal) that any collision will shoot the particle out of that element. That way, R is the number of collisions in d^3rd^3v. For $\overline{R}$ we can interpret it as the number of collisions outside the original element that end up in d^3rd^3v, i.e., number of collisions whose final state is in d^3rd^3v.

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Remark 1.4.1 Testing

$$a = b \tag{1.5}$$

1.5 Exercises